

Md. Sajib Hossain

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RESEARCH PROFILE

- Currently a Research Scholar at the University of Rhode Island's Wearable Biosensing Lab, working on sensor design, fabrication, signal processing, and ML model development.
- Four years of research in functional nanomaterials, electrochemistry, and conductive composite fabrication directly relevant to electrocatalysis research.
- Synthesized and characterized battery/energy-relevant nanomaterials: MXene ($\text{Ti}_3\text{C}_2\text{T}_x$), graphene oxide, g- C_3N_4 , silver nanowires, and zinc oxide.
- Hands-on electrochemical and electrical characterization: LCR meters, electrometers, oscilloscopes, conductivity analysis, and device fabrication.
- 5 peer-reviewed publications and 2 Indian patent co-inventorships; ICCR Scholar (IIT Delhi).

CURRENT POSITION

Research Scholar — University of Rhode Island, RI, USA **2026 – Present**

Department of Electrical, Computer & Biomedical Engineering | Wearable Biosensing Lab

- Sensor design, fabrication, and analysis for wearable biosensing applications.
- Device fabrication and bench-level testing of biosensor prototypes.
- Signal processing of physiological/sensor data for biomedical applications.
- Development of machine learning models for sensor data analysis and prediction.
- Preparation and management of IRB protocols for human-subjects research studies.

EDUCATION

Indian Institute of Technology (IIT) Delhi, India **Jul 2023 – May 2025**

MTech in Textile Chemical Processing | CGPA: 8.32 / 10

Thesis: Multifunctional MXene/PEDOT:PSS screen-printed conductive fabric for sensor, Joule heating, and EMI shielding

University of Dhaka, Bangladesh **Feb 2018 – May 2023**

Bachelor of Textile Engineering | CGPA: 3.53 / 4.0

RESEARCH EXPERIENCE

Indian Institute of Technology Delhi, Master's Research **Jul 2023 – May 2025**

Project 1: Multifunctional Conductive Fabric — Electrocatalytic & Electrochemical Characterization

- Synthesized MXene ($\text{Ti}_3\text{C}_2\text{T}_x$) nanosheets and formulated rheologically optimized MXene/PEDOT:PSS screen-printing pastes for conductive fabric fabrication.
- Engineered 4-layer conductive architecture achieving $20\ \Omega/\text{sq}$ sheet resistance; conductivity retained over 25 laundering cycles, demonstrating electrochemical stability.
- Characterized electrochemical behavior and electrical properties using LCR meters, electrometers, and oscilloscopes — skills directly transferable to electrocatalyst electrode characterization.
- Filed Indian patent (App. No. 202611020096); manuscript in preparation.

Project 2: MXene-Modified Biopolymer Nanocomposites — Ion-Exchange & Interface Study

- Synthesized amphoteric MXene/sodium alginate/chitosan nanocomposites; analyzed ion-exchange and surface adsorption mechanisms for anionic and cationic species.

- 12 adsorption-desorption cycles demonstrated MXene surface stability, providing direct insight into electrode-electrolyte interface chemistry.
- Manuscript under submission.

Project 3: Graphitic Carbon Nitride (g-C₃N₄) Synthesis & Charge-Separation Optimization

- Synthesized and optimized g-C₃N₄ via thermal polymerization; systematically tuned surface area, light absorption, and charge-separation efficiency relevant to electrocatalytic applications.
- Results applied to UV-shielding films; published in ACS Applied Materials & Interfaces (2023).

IIT Delhi – Research Intern

Mar 2023 – Jul 2023

- Synthesized g-C₃N₄ and fabricated alginate/GCN nanocomposites; characterized via XRD, TGA, UTM, and UV–Vis spectroscopy.

Gopalganj Science & Technology University – Research Intern

Jun 2021 – Oct 2021

- Synthesized graphene oxide-based composite adsorbents; characterized structure and assessed electrochemical dye-adsorption properties.

TECHNICAL SKILLS

Biosensing & Computational: Wearable sensor design and fabrication, signal processing, machine learning model development, IRB protocol preparation.

Nanomaterial Synthesis: MXene (Ti₃C₂T_x), Graphene oxide, Graphitic Carbon Nitride (g-C₃N₄), Silver nanowires (AgNW), Zinc oxide, polymer nanocomposites, biopolymer hydrogels.

Electrochemical & Electrical: LCR meter, electrometer, oscilloscope, electrical conductivity analysis, electrochemical behavior characterization, device fabrication.

Structural & Thermal: XRD, SAXS, TGA, DSC, DMTA, Instron/UTM.

Surface & Morphological: SEM/EDX, TEM, BET, 3D Optical profilometer, contact angle goniometer, zeta potential.

Spectroscopy: FT-IR (ATR & KBr), UV–Vis, NMR, mass spectrometry.

Processing: Screen printing, spray/spin/dip/doctor-blade coating, extrusion, rheometry.

Software: Origin Pro, TA Analysis, ChemDraw, Avogadro, Overleaf, Mendeley, Photoshop; Python (developing).

PUBLICATIONS

Peer-Reviewed Journals

- Hossain, M. S. et al. Prolonging Fruit Shelf-Life with GCN-Incorporating Biopolymer Composite Coating. Sustainable Food Technol. 2026. DOI: 10.1039/sftb00744e.
- Hossain, M. S.; Adak, B.; Khushbu; Mukhopadhyay, S. g-C₃N₄-Reinforced Chitosan/Ca²⁺-Alginate Nanocomposite Films for Sustainable Packaging. ACS Appl. Bio Mater. 2025. DOI: 10.1021/acsabm.5c01293.
- Islam, M.; Hossain, M. S. (co-first author) et al. Recent Advancements in Carbon-Based Composites as Electrodes for High-Performance Supercapacitors. J. Energy Storage 2025, 107, 114838.
- Hasan, M. M.; Hossain, M. S. et al. Nontoxic Cross-Linked g-C₃N₄-Alginate Biodegradable Films for UV and Blue Light Shielding. ACS Appl. Mater. Interfaces 2023, 15, 32011–32023.
- Sarker Ratna, A.; Hossain, S. et al. Corn Husk Cellulose/PVA Bio-Composite Films with ZnO Nanoparticles. Bioresour. Technol. Reports 2023, 23, 101570.

Manuscripts Under Submission / In Preparation

- Hossain, M. S. et al. Flexible Multifunctional Fabric-Based Sensor and Joule Heating for Healthcare. (In preparation)

- Hossain, M. S. et al. MXene-Modified Sodium Alginate/Chitosan Amphoteric Nanocomposites as Efficient Adsorbent for Anionic and Cationic Dyes. (Under submission)

PATENT APPLICATIONS

- Titanium Carbide Flake-Based Screen-Printed Washable Fabric for EMI/RF Shielding. Indian Patent App. No. 202611020096. Co-inventor (PI: Dr. S. Mukhopadhyay, IIT Delhi).
- Biobased Porous Beads for Textile Wastewater Treatment. Indian Patent Application. Co-inventor (PI: Dr. S. Mukhopadhyay, IIT Delhi).

INDUSTRY EXPERIENCE

Archroma International Pvt. Ltd., Vadodara, India

Jun 2024 – Jul 2024

- Optimized Tubular and Spiral Reverse Osmosis membrane process parameters; conducted end-to-end process analysis using WaveRO software.
- Assessed chemical compatibility and cleaning chemical selection for membrane systems.

CONFERENCES & PRESENTATIONS

- Best Poster Award — Indo-Australian Conference on Biomimicry, Jamia Millia Islamia, India (Feb 2025).
- Poster: Biodegradable GCN-Biopolymer Coating for Fruit Shelf-Life — International Conclave on Materials, Energy & Climate, University of Dhaka (Dec 2022).
- Poster: Eco-Friendly Textile from Corn Husk — 2nd Int'l Conference on Functional Textiles & Clothing, IIT Delhi (2020).
- Organizer, 7th International Conference on Technical Textiles and Nonwovens (ICTN), India (2023).

AWARDS & HONORS

- Indian Council for Cultural Relations (ICCR) Scholar — India (2023).
- Champion, NASA Space Apps Challenge — Bangladesh (2019); Global Nominee (2021).
- National General Scholarship — Bangladesh (2011).

LEADERSHIP

Founder & President — Niter Science Society

2022 – 2023

- Founded student scientific society; led strategic planning, academic events, and external institutional collaborations.

Convenor — BASIS Students Forum, Bangladesh

2020 – 2022

- Coordinated ICT initiatives with Bangladesh Association of Software & Information Services; organized workshops and industry networking events.